In [3]:
```
%display latex
```

Problem 7

In [4]:
```
#vector v
v = vector([3,2])
```

In [5]:
```
#matrix that rotate the vector counterclokwise by 15 degrees (15 = 180/12 =
π/12)
a=pi/12
A = matrix(SR,[[cos(a), -sin(a)],
               [sin(a),  cos(a)]])
A
```

Out[5]:
$$\begin{pmatrix} \frac{1}{4}\sqrt{6}+\frac{1}{4}\sqrt{2} & -\frac{1}{4}\sqrt{6}+\frac{1}{4}\sqrt{2} \\ \frac{1}{4}\sqrt{6}-\frac{1}{4}\sqrt{2} & \frac{1}{4}\sqrt{6}+\frac{1}{4}\sqrt{2} \end{pmatrix}$$

In [6]:
```
#matrix that rotate the vector counterclokwise by 45 degrees (45 = π/4)
b=pi/4
B = matrix(SR,[[cos(b), -sin(b)],
               [sin(b),  cos(b)]])
B
```

Out[6]:
$$\begin{pmatrix} \frac{1}{2}\sqrt{2} & -\frac{1}{2}\sqrt{2} \\ \frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \end{pmatrix}$$

In [7]:
```
#compute the rotated vector vA
vA = A*v
vA
```

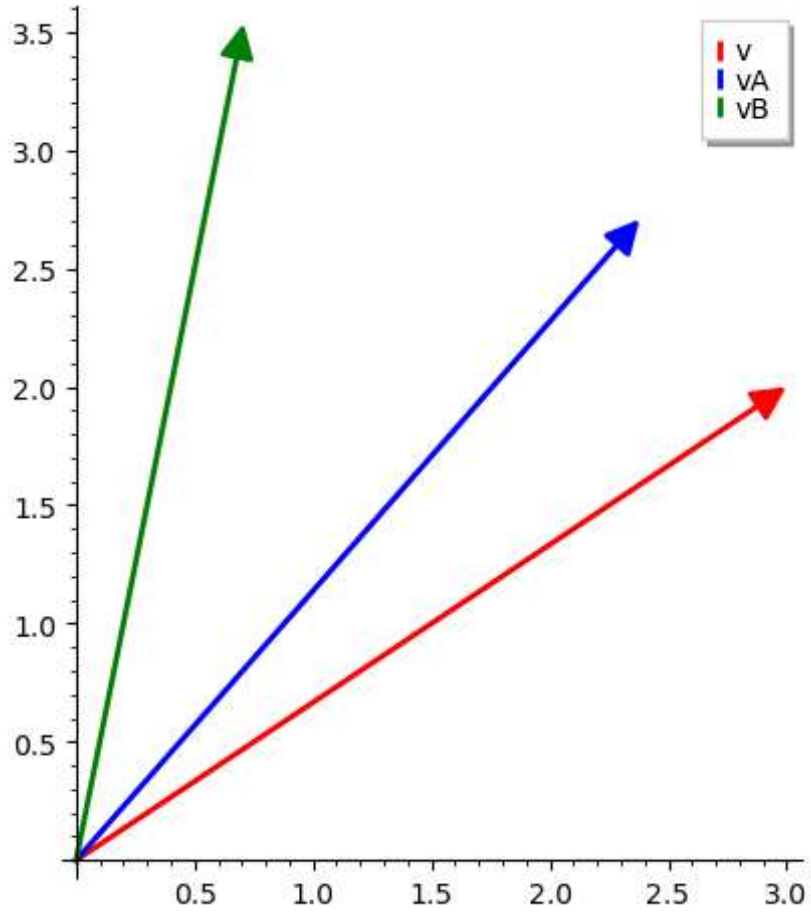
Out[7]:
$$\left(\frac{1}{4}\sqrt{6}+\frac{5}{4}\sqrt{2}, \frac{5}{4}\sqrt{6}-\frac{1}{4}\sqrt{2}\right)$$

In [8]:
```
#compute the rotated vector vB
vB = B*v
vB
```

Out[8]:
$$\left(\frac{1}{2}\sqrt{2}, \frac{5}{2}\sqrt{2}\right)$$

In [9]:
```
#sketch vectors vA, vB and v on a coordinate grid
origin = vector([0,0])
Arrows = arrow(origin, v, color='red', width=2, legend_label='v') +\
arrow(origin, vA, color='blue', width=2, legend_label='vA') +\
arrow(origin, vB, color='green', width=2, legend_label='vB')
Arrows.show(aspect_ratio=1)
```

Out[9]:



In [10]:

```
#compare the lenght of vectors vA, vB and v
Vnorm = v.norm().simplify_full()
VAnorm = vA.norm().simplify_full()
VBnorm = vB.norm().simplify_full()
Vnorm, VAnorm, VBnorm
```

Out[10]: $\left(\sqrt{13}, \sqrt{13}, \sqrt{13}\right)$

Comment:

Matrices A and B are rotation matrices. Any rotation matrix is orthogonal and preserve the length of vectors. That's why the lengths of v, vA, and vB are equal.

In [11]:

```
#find the angle (in degrees) between vA and vB
angle = acos((vA.dot_product(vB)) / (vA.norm()*vB.norm()))
round(float(angle*180/pi))
```

Out[11]: 30.0

Problem 8

```
#vectors
u1 = vector([1,0,2,1,3])
u2 = vector([0,1,1,2,1])
u3 = vector([2,1,0,1,4])
```

```
#matrix A=[u1,u2,u3]
A = column_matrix([u1,u2,u3])
A
```

Out[13]:

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & 1 \\ 3 & 1 & 4 \end{pmatrix}$$

```
#check whether the vectors u1, u2, u3 are linearly independent
if A.rank() == 3:
    print ('Vectors v1,v2 and v3 are linearly independent')
else:
    print ('Vectors are linearly dependent')
```

Out[14]: Vectors v1,v2 and v3 are linearly independent

```
#find the projection matrix P that maps R^5 onto U
P = A * (A.transpose()* A).inverse() * A.transpose()
P
```

Out[15]:

$$\begin{pmatrix} \frac{15}{56} & \frac{1}{56} & -\frac{9}{56} & -\frac{1}{8} & \frac{11}{28} \\ \frac{1}{56} & \frac{23}{56} & -\frac{15}{56} & \frac{23}{56} & \frac{1}{28} \\ -\frac{9}{56} & -\frac{15}{56} & \frac{47}{56} & \frac{9}{56} & \frac{3}{28} \\ -\frac{1}{8} & \frac{23}{56} & \frac{9}{56} & \frac{39}{56} & \frac{1}{28} \\ \frac{11}{28} & \frac{1}{28} & \frac{3}{28} & \frac{1}{28} & \frac{11}{14} \end{pmatrix}$$

```
#find the projection of the vector x onto U
x = vector([3,-1,4,0,2])
Px = P * x
Px
```

Out[16]:

$$\left(\frac{13}{14}, -\frac{19}{14}, \frac{47}{14}, -\frac{1}{14}, \frac{22}{7}\right)$$

```
#compute the projection error e=x-Px
e = x - Px
e
```

Out[17]:

$$\left(\frac{29}{14}, \frac{5}{14}, \frac{9}{14}, \frac{1}{14}, -\frac{8}{7}\right)$$

```
#verify that e is orthogonal to U
if (e.dot_product(u1), e.dot_product(u2), e.dot_product(u3)) == (0,0,0):
    print('e is orthogonal to U')
else:
    print('e is not orthogonal to U')
(e.dot_product(u1), e.dot_product(u2), e.dot_product(u3))
```

Out[18]: e is orthogonal to U

$$(0, 0, 0)$$

```
#compute norm of e
enorm = e.norm()
enorm
```

Out[19]: $\sqrt{\frac{43}{7}}$

Problem 9

$\langle x, y\rangle = x^T A y$, $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 1 & k \end{bmatrix}$, $k \in \mathbb{R}$, $x = (x_1, x_2, x_3)^T$
$y = (y_1, y_2, y_3)^T$

a) For $\langle x, y\rangle = x^T A y$ to define an inner product on $\mathbb{R}^3$, the matrix A must be symmetric and positive definite.

i) $A^T = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 1 & k \end{bmatrix} = A$: is symmetric

ii) A is positive definite if and only if all its leading principal minors are positive.

$\Delta_1 = a_{11} = 2 > 0$

$\Delta_2 = \det \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = 2 \cdot 2 - (-1)(-1) = 4 - 1 = 3 > 0$

$\Delta_3 = \det(A) = 2(2k - 1 \cdot 1) - (-1)(-1 \cdot k - 1 \cdot 0) + 0(-1 \cdot k - 1 \cdot 0) = 4k - 2 - k = 3k - 2$

$3k - 2 > 0$

$3k > 2$

$k > 2/3$

All principal minors are positive when $k > 2/3$

Thus A defines an inner product if and only if $k > 2/3$

b) $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ - basis vectors.

Norm of a standard basis vector relative to symmetric positive definite matrix A is a square root of the corresponding diagonal entry of A : $\|e_i\|_A = \sqrt{A_{ii}}$

$\|e_1\| = \sqrt{A_{11}} = \sqrt{2}$

$\|e_2\| = \sqrt{A_{22}} = \sqrt{2}$

$\|e_3\| = \sqrt{A_{33}} = \sqrt{k}$

c) $x=(1,0,1)^T$, $y=(0,1,1)^T$

The distanse between x and y is $d(x,y)=\|x-y\|$

$$x-y=\begin{pmatrix}1\\0\\1\end{pmatrix}-\begin{pmatrix}0\\1\\1\end{pmatrix}=\begin{pmatrix}1-0\\0-1\\1-1\end{pmatrix}=\begin{pmatrix}1\\-1\\0\end{pmatrix}$$

$$\langle x-y,x-y\rangle=(x-y)^T A(x-y)=[1\ -1\ 0]\cdot\begin{bmatrix}2&-1&0\\-1&2&1\\0&1&k\end{bmatrix}\cdot\begin{bmatrix}1\\-1\\0\end{bmatrix}=$$

$$=[1\ -1\ 0]\cdot\begin{bmatrix}2\cdot1+(-1)(-1)+0\cdot0\\-1\cdot1+2(-1)+1(-1)\\0\cdot1+1(-1)+k\cdot0\end{bmatrix}=[1\ -1\ 0]\cdot\begin{bmatrix}3\\-3\\-1\end{bmatrix}=1\cdot3+(-1)(-3)+0(-1)=$$

$=3+3=6$

$d(x,y)=\|x-y\|=\sqrt{\langle x-y,x-y\rangle}=\boxed{\sqrt{6}}$

d) $\langle e_1,e_2\rangle_A=e_1^T A e_2=A_{12}=-1\neq0 \Rightarrow e_1$ and e_2 are not orthogonal

Let's use the Graw-Schmidt process:

for $u_1=e_1=(1,0,0)^T$

$$u_2=e_2-\frac{\langle e_2,u_1\rangle}{\langle u_1,u_1\rangle}u_1$$

$\langle e_2,u_1\rangle=\langle e_2,e_1\rangle=\langle e_1,e_2\rangle=-1$

$\langle u_1,u_1\rangle=\langle e_1,e_1\rangle=A_{11}=2$

$$u_2=\begin{bmatrix}0\\1\\0\end{bmatrix}-\frac{-1}{2}\begin{bmatrix}1\\0\\0\end{bmatrix}=\begin{bmatrix}0\\1\\0\end{bmatrix}+\begin{bmatrix}1/2\\0\\0\end{bmatrix}=\begin{bmatrix}1/2\\1\\0\end{bmatrix}$$

$$\boxed{u_1=(1,0,0)^T,\ u_2=(1/2,1,0)^T}$$

Problem 10

$v_1=(2,1,1,1)^T$, $v_2=(1,0,1,1)^T$, $v_3=(-2,-1,0,1)^T$

$W=\text{span}\{v_1,v_2,v_3\}$ $\qquad u=(6,3,9,6)^T$

a) If x is orthogonal to v_1, v_2 and v_3, then by linearity of the inner product, x is orthogonal to any vector in W

$$x\in W^\perp \iff \langle x, v_i\rangle = 0$$

For $x=(x_1,x_2,x_3,x_4)^T$

$v_1x = 2x_1+x_2+x_3+x_4=0$

$v_2x = x_1+x_3+x_4=0$

$v_3x = -2x_1-x_2+x_4=0$

i) $v_1x+v_3x = 2x_1+x_2+x_3+x_4+(-2x_1)+(-x_2)+x_4 = x_3+2x_4=0 \Rightarrow x_3=-2x_4$

ii) $v_2x = x_1+x_3+x_4 = x_1+(-2x_4)+x_4 = x_1-x_4=0 \Rightarrow x_1=x_4$

iii) $v_1x = 2x_1+x_2+x_3+x_4 = 2(x_4)+x_2+(-2x_4)+x_4 = x_2+x_4=0 \Rightarrow x_2=-x_4$

So every solution is $x_4(1,-1,-2,1)^T$

Therefore $W^\perp=\text{span}\{(1,-1,-2,1)^T\}$, $w_0=(1,-1,-2,1)^T$

$\text{Dim}\,W^\perp=1 \Rightarrow \dim W=3$

b) We can project u onto W subtracting its projection onto $W^\perp$, because $\mathbb{R}^4=W+W^\perp$

i) Check if v_1, v_2, v_3 are linearly independent

$av_1+bv_2+cv_3=0$

1) $2a+1b+(-2)c=0$ $\qquad 2a+b-2c=0$

2) $1a+0b+(-1)c=0$ $\qquad a-c=0 \Rightarrow a=c$

3) $1\cdot a+1\cdot b+0c=0$ $\qquad a+b=0 \Rightarrow a=-b$

4) $1\cdot a+1\cdot b+1\cdot c=0$ $\qquad a+b+c=0 \Rightarrow a+(-a)+a=0 \Rightarrow a=0$

$\Rightarrow b=0, c=0$ - linearly independent

ii) Projection of u onto w_0

$$u_\perp = \frac{u\cdot w_0}{w_0\cdot w_0}w_0$$

$u\cdot w_0=(6,3,9,6)\cdot(1,-1,-2,1)=6\cdot1+3(-1)+9(-2)+6\cdot1=6-3-18+6=-9$

$$w_0 \cdot w_0 = (1,-1,-2,1)(1,-1,-2,1) = 1^2 + (-1)^2 + (-2)^2 + 1^2 = 1+1+4+1 = 7$$

$$u_\perp = \frac{u \cdot w_0}{w_0 \cdot w_0} w_0 = \frac{-9}{7}(1,-1,-2,1) = \left(-9/7, \frac{9}{7}, \frac{18}{7}, \frac{-9}{7}\right)^T$$

iii) Projection of u onto W:

$$u_W = u - u_\perp = (6,3,9,6)^T - \left(\frac{-9}{7}, \frac{9}{7}, \frac{18}{7}, \frac{-9}{7}\right)^T = \left(6+\frac{9}{7}, 3-\frac{9}{7}, 9-\frac{18}{7}, 6+\frac{9}{7}\right)^T =$$

$$= \left(\frac{42}{7}+\frac{9}{7}, \frac{21}{7}-\frac{9}{7}, \frac{63}{7}-\frac{18}{7}, \frac{42}{7}+\frac{9}{7}\right)^T = \boxed{\left(\frac{51}{7}, \frac{12}{7}, \frac{45}{7}, \frac{51}{7}\right)^T = u_W}$$

c) $e = u - u_W = u_\perp = \left(-\frac{9}{7}, \frac{9}{7}, \frac{18}{7}, -\frac{9}{7}\right)^T$

$$\|e\| = \sqrt{\left(-\frac{9}{7}\right)^2 + \left(\frac{9}{7}\right)^2 + \left(\frac{18}{7}\right)^2 + \left(-\frac{9}{7}\right)^2} = \sqrt{\frac{81}{49} + \frac{81}{49} + \frac{324}{49} + \frac{81}{49}} = \sqrt{\frac{567}{49}} = \sqrt{\frac{81}{7}} = \frac{9}{\sqrt{7}}$$

$$\boxed{\|e\| = \frac{9}{\sqrt{7}}}$$